Coincidence of perfect-matching, connectivity and minimum-degree hitting times in product graphs

Sahar Diskin, Anna Geisler

arxiv:2404.14020v4; CC BY 4.0

Editorial note. Original complete argument retained with version/license/figure dependencies; all support belongs to this single counted target.

Editorial note. The prepared deterministic dimension threshold and public nonbipartite-ge lemma are proper support; neither provides the random hitting-time theorem. The immediately stronger window formulation is support, not a second counted target.

Allowed standard prerequisites (editorial).

- Standard subject prerequisites. Standard named subject results cited by the authors may be used; all paper-specific supporting arguments are retained verbatim. No modified assumptions or replacement proof.

1 Introduction

1.1 Background and main results

Given two graphs $H_1 = (V_1, E_1)$ and $H_2 = (V_2, E_2)$, the *Cartesian product* $H = H_1 \square H_2$ is the graph whose vertex set is $V_1 \times V_2$, and for $u_1, v_1 \in V_1$ and $u_2, v_2 \in V_2$, we have that $\{(u_1, u_2), (v_1, v_2)\} \in E(H)$ either if $u_1 = v_1$ and $\{u_2, v_2\} \in E(H_2)$ or if $u_2 = v_2$ and $\{u_1, v_1\} \in E(H_1)$. More generally, given t graphs, $H_i, \dots, H_t$, their Cartesian product $G = \square_{i=1}^{t} H_i$ is the graph with the vertex set

$$V \coloneqq \{v = (v_1, \dots, v_t) \colon v_i \in V(H_i) \text{ for all } i \in [t]\},$$

and the edge set

$$\left\{ uv \colon \begin{array}{c} \text{there is some } i \in [t] \text{ such that } \{u_i, v_i\} \in E(H_i) \\ \text{and } u_j = v_j \text{ for all } j \neq i \end{array} \right\}.$$

We call $H_1, H_2, \dots, H_t$ the *base graphs of* G. Cartesian product graphs arise naturally in many contexts and have received much attention in combinatorics, probability, and computer science. Many classical graphs, which have been extensively studied, are in fact Cartesian product graphs: the t-dimensional torus is the Cartesian product of t copies of the cycle C_k, the t-dimensional grid is the Cartesian product of t copies of the path P_k, and the binary t-dimensional hypercube Q^t is the Cartesian product of t copies of a single edge K_2. We refer the reader to [18] for a systematic treatment of Cartesian product graphs and related graph products.

A *perfect matching* in a graph G is a collection of vertex disjoint edges which cover the vertex set of G. A matching is *nearly-perfect* if it leaves exactly one vertex unmatched. In this paper, we study perfect matchings of high-dimensional product graphs, both in the deterministic and in the random setup.

We begin with the deterministic part. Observe that if one base graph, say H_j, has a perfect matching, then $G = \square_{i=1}^{t} H_i$ has a perfect matching: for each fixed choice of the other coordinates we may take the corresponding copy of that matching in the j-th coordinate. Kotzig [21] proved a stronger statement: if all base graphs are regular and either one base graph has a 1-factorisation, or two base graphs have perfect matchings, then the product has a 1-factorisation (recall that a 1-factorisation is a decomposition of the edge set into perfect matchings). However, when none of the base graphs has a perfect matching, the existence of a perfect matching in the product is not automatic.

Nevertheless, for any two graphs, the proportion of vertices covered by a maximum matching in their product is at least the corresponding proportion in either base graph: one simply takes a maximum matching in that base graph for every fixed choice of the other coordinate. It is therefore natural to ask: can one take sufficiently many products, that is, consider a product graph of high enough dimension, to force a perfect or nearly-perfect matching? If so, what is an optimal bound on the required dimension? Our first result answers the first question in the positive, and gives a *tight* bound on the dimension of a Cartesian product graph that ensures the existence of a perfect matching.

Theorem 1. *Let $M \geq 2$ and let $t > 2\ln M$ be integers. For every $i \in [t]$, let H_i be a connected d_i-regular graph with $1 < |V(H_i)| \leq M$, and write $G = \square_{i=1}^t H_i$. Then G has a (nearly-)perfect matching.*

Furthermore, there is a sequence of such graphs $(H_i)_{i\in[t]}$ such that for $t < \ln M/3$ the product $G = \square_{i=1}^t H_i$ does not have a perfect matching.

In the following remark, we show that some regularity assumption is necessary for a perfect matching to appear.

Remark 1.1. *Let $G = \square_{i=1}^t K_{1,s}$. The graph G is bipartite; writing $\mathcal{E}$ and $\mathcal{O}$ for the sides according to the parity of the number of leaf coordinates, we have*

$$|\mathcal{E}| = \sum_{\substack{0\leq k\leq t\\ k \text{ even}}} \binom{t}{k} s^k, \qquad |\mathcal{O}| = \sum_{\substack{0\leq k\leq t\\ k \text{ odd}}} \binom{t}{k} s^k.$$

Consequently,

$$\big||\mathcal{E}| - |\mathcal{O}|\big| = |1-s|^t.$$

For $s \geq 3$ this imbalance is larger than one, so G has neither a perfect nor a nearly-perfect matching.

We now turn to the random part. Given a graph G and probability p, we form the percolated random subgraph $G_p \subseteq G$ by including every edge of G independently with probability p (note that $G(n,p)$ is then $(K_n)_p$).

The study of Q_p^t was initiated by Sapoženko [25] and by Burtin [7], who showed that the (sharp) threshold for connectivity is $p^* = \frac{1}{2}$: when $p < \frac{1}{2}$ is a constant, with high probability (whp), that is, with probability tending to one as t tends to infinity, Q_p^t is disconnected, whereas for $p > \frac{1}{2}$ constant, whp Q_p^t is connected. Erdős and Spencer [15] conjectured that Q_p^t undergoes a phase transition with respect to its component structure, that is, the typical emergence of a giant component (a connected component containing a linear fraction of the vertices) around $p = \frac{1}{t}$, similar to that of $G(n,p)$ around $p = \frac{1}{n}$. This conjecture was confirmed by Ajtai, Komlós, and Szemerédi [1], with subsequent work by Bollobás, Kohayakawa, and Łuczak [5]. We refer the reader to [22] for a modern short proof of this result.

In recent years, there has been an effort to generalise these results to a wider family of product graphs. Lichev [23] gave sufficient conditions, in terms of the base graphs, for the typical emergence of a giant component in bond percolation on high-dimensional product graphs. Diskin, Erde, Kang, and Krivelevich [10] improved upon this, giving sufficient and tight conditions for the typical emergence of a giant component. Furthermore, they showed that assuming the base graphs are regular, one can give a rather precise description, similar to that in $G(n,p)$, both of the typical component structure [11], and of the asymptotic combinatorial properties of the giant component [9].

Returning to the dense regime, that is, when p is constant, connectivity and the existence of a perfect matching in Q_p^t have been studied in detail. In particular, Bollobás [4] obtained a *hitting time* result for the *random graph process* on Q^t. Given a graph Γ, the random graph process on Γ is defined as a random sequence of nested graphs $\Gamma(0) \subseteq \ldots \subseteq \Gamma(|E(\Gamma)|)$ together with an ordering σ on $E(\Gamma)$, chosen uniformly at random from among all $|E(\Gamma)|!$ such orderings. We set $\Gamma(0)$ to be the empty graph on $V(\Gamma)$. Given $\Gamma(i)$, with $0 \leq i < |E(\Gamma)|$, we form $\Gamma(i+1)$ by adding the $(i+1)$-st edge, according to the ordering σ, to $\Gamma(i)$. The hitting time of a monotone increasing, non-empty graph property $\mathcal{P}$, is the random variable equal to the index τ for which $\Gamma(\tau) \in \mathcal{P}$, but $\Gamma(\tau-1) \notin \mathcal{P}$. Note that having minimum degree one, connectivity, and the existence of a perfect matching are all monotone increasing properties. Furthermore, observe that for a graph Γ to be connected or to contain a perfect matching, the minimum degree of Γ has to be at least one. A classical result of Erdős and Rényi [14], and of Bollobás and Thomason [6], is that in the random process on K_n, whp the hitting time for minimum degree one, connectivity, and the existence of a perfect matching are the same. In 1990, Bollobás [4] showed that the same phenomenon holds in the random graph process on Q^t. Subsequent work by Joos determined

the threshold for connectivity for Cartesian powers of graphs [19, 20], that is, for $G = \square_{i=1}^{t} H$ where all the base graphs $H_i = H$ are the same.

The corresponding statement for Cartesian products whose base graphs have bounded order is our main probabilistic result. Here and below, an event holds *with high probability*, abbreviated whp, if its probability tends to one as $t \to \infty$ with M fixed.

Theorem 2. *Let $M \geq 2$ be fixed, and for every $i \in [t]$ let H_i be a connected d_i-regular graph with $1 < |V(H_i)| \leq M$. Let $G = \square_{i=1}^{t} H_i$, and suppose $|V(G)|$ is even. In the random graph process on G, let τ_1, τ_2, and τ_3 be the hitting times for minimum degree one, connectivity, and the existence of a perfect matching, respectively. Then, whp, $\tau_1 = \tau_2 = \tau_3$.*

We use a standard coupling of the random graph process with the random subgraph formed by percolation: assign to every $e \in E(G)$ a label ξ_e that is uniform on $[0, 1]$, and set

$$G_p = \big(V(G), \{e \in E(G) : \xi_e \leq p\}\big).$$

The labels are almost surely distinct, and $(G_p)_{0 \leq p \leq 1}$ then naturally couples with the random graph process. For a graph Γ, write $\mathrm{iso}(\Gamma)$ for the number of its isolated vertices. We work on the following interval around the disappearance of the last isolated vertices. Write $n = |V(G)|$ and $d = d(G)$, let $\omega = \omega(d)$ satisfy

$$\omega \to \infty \qquad \text{and} \qquad \omega \leq d^{1/100},$$

and define p_- and p_+ by

$$(1 - p_-)^d = \frac{\omega}{n}, \qquad (1 - p_+)^d = \frac{1}{\omega n}. \tag{1}$$

We are now ready to state the following structural result for random subgraphs of G.

Theorem 3. *Let $M \geq 2$ be fixed, and for every $i \in [t]$ let H_i be a connected d_i-regular graph with $1 < |V(H_i)| \leq M$. Let $G = \square_{i=1}^{t} H_i$, and suppose $|V(G)|$ is even. Let $p \in [p_-, p_+]$. Then, whp, the following statements hold simultaneously.*

(a) In G_{p_-} there is a unique component containing every non-isolated vertex. The isolated vertices are pairwise at distance at least three in G, and

$$1 \leq \mathrm{iso}(G_{p_-}) \leq \sum_{i=1}^{t} d_i.$$

(b) The graph G_{p_+} has no isolated vertices.

(c) For any $p \in [p_-, p_+]$, if G_p has minimum degree at least one, then it has a perfect matching.

Note, that Theorem 3 readily implies Theorem 2: Part (a) and (b) of Theorem 3 imply that with high probability $\tau_1 \in [p_-, p_+]$ and G_{τ_1} is connected, so $\tau_2 = \tau_1$. Since $\delta(G_{\tau_1}) \geq 1$, part (c) of Theorem 3 gives a perfect matching in G_{τ_1}. Having a perfect matching is an increasing property and implies minimum degree at least one; therefore $\tau_3 = \tau_1$ as well.

One of the key challenges in general product graphs, as compared to the hypercube, is the lack of bipartitedness. In particular, the existence of a perfect matching cannot be proved by verifying Hall's condition; instead, we use the Tutte–Berge formula and the Edmonds-Gallai decomposition.

Structure of the paper The paper is structured as follows. In Section 2, we collect some notation and results that will be of use throughout the paper. In Section 3 we give a proof outline containing the key ideas. In Section 4 we cover the deterministic setting and prove Theorem 1. Switching to the random setting, we treat connectivity in Section 5, and the existence of a perfect matching in Section 6. Finally, we conclude with a discussion and some open problems in Section 7.

2 Preliminaries

Notation and definitions. Throughout, $M \geq 2$ is fixed, $(H_i)_{i=1}^{t}$ is a sequence of connected d_i-regular graphs with $1 < |V(H_i)| \leq M$, and

$$G = (V, E) = \square_{i=1}^{t} H_i, \qquad n = |V|, \qquad d = d(G) = \sum_{i=1}^{t} d_i.$$

We call t the *dimension* of G, and, for $u = (u_1, \dots, u_t) \in V$, we call u_i the i-th coordinate of u. Since $2^t \le n \le M^t$ and $t \le d \le (M-1)t$, we have $d = \Theta_M(\ln n)$. More precisely, we will use repeatedly that $\ln n \ge \frac{\ln 2}{M-1} d$. For a graph Γ and a set $A \subseteq V(\Gamma)$, write $E_\Gamma(A)$ for the set of edges of Γ with both endpoints in A, and set $e_\Gamma(A) \coloneqq |E_\Gamma(A)|$. For disjoint sets $A, B \subseteq V(\Gamma)$, write $E_\Gamma(A,B)$ for the set of edges of Γ between A and B, and set $e_\Gamma(A,B) \coloneqq |E_\Gamma(A,B)|$. We abbreviate $A^c = V(\Gamma) \setminus A$ when the graph in question is clear. Furthermore, we denote by $\lambda(\Gamma)$ the minimum number of edges whose removal disconnects Γ and call this the edge-connectivity of Γ. The deficit of Γ, abbreviated by $\mathrm{def}(\Gamma)$, is the number of vertices left unmatched by a maximum matching. Finally, we write $\mathrm{odd}(\Gamma)$ for the number of odd-order components in a graph Γ.

External results We use the following isoperimetric inequalities for product graphs with bounded-order base graphs; they follow from Theorems 1 and 2 of [9] and Section 3.4 of [12].

Theorem 2.1. *Let $M \ge 2$ be fixed, and for every $i \in [t]$ let H_i be a connected d_i-regular graph with $1 < |V(H_i)| \le M$. Let $G = (V,E) = \square_{i=1}^t H_i$. For every non-empty $A \subseteq V$,*

$$e_G(A, A^c) \ge \frac{e}{M} |A| \ln \frac{n}{|A|}, \tag{2}$$

$$e_G(A, A^c) \ge |A| \big(d - (M-1) \log_M |A|\big). \tag{3}$$

We also use the standard rooted-tree bound (see, for example, [3, Lemma 2]).

Lemma 2.2. *Let Γ be a graph of maximum degree at most d, let $k \ge 1$ be an integer, and fix some $v \in V(\Gamma)$. The number of k-vertex trees in Γ containing v is at most $(ed)^{k-1}$.*

The Tutte-Berge formula The Tutte-Berge formula [2] is our first tool in order to treat the existence of perfect matchings. It implies that the number of unmatched vertices in a maximum matching in a graph Γ is equal to

$$\mathrm{def}(\Gamma) = \max_{U \subseteq V(\Gamma)} \{\mathrm{odd}\,(\Gamma[V(\Gamma) \setminus U]) - |U|\}, \tag{4}$$

where we recall that $\mathrm{odd}\,(\Gamma[V(\Gamma) \setminus U])$ is the number of connected components with an odd number of vertices in $\Gamma[V(\Gamma) \setminus U]$. In particular, a graph Γ has a perfect matching if and only if for every subset $U \subseteq V(\Gamma)$, the subgraph $\Gamma[V(\Gamma) \setminus U]$ has at most $|U|$ connected components with an odd number of vertices. Note here that if Γ has isolated vertices, then choosing $U = \varnothing$ witnesses that there is no perfect matching in Γ.

The Gallai-Edmonds decomposition The second tool provides us with a structural description of any maximum matching in Γ. We define

$$\begin{aligned} D_\Gamma &\coloneqq \{v \in V(\Gamma) : \text{some maximum matching of } \Gamma \text{ leaves } v \text{ unmatched}\}, \\ A_\Gamma &\coloneqq N_\Gamma(D_\Gamma) \setminus D_\Gamma, \\ C_\Gamma &\coloneqq V(\Gamma) \setminus (D_\Gamma \cup A_\Gamma). \end{aligned}$$

The uniquely determined partition

$$V(\Gamma) = D_\Gamma \sqcup A_\Gamma \sqcup C_\Gamma$$

is the *Gallai–Edmonds decomposition* of Γ. A graph is called *factor-critical* if deleting any one of its vertices leaves a graph with a perfect matching. Writing m_γ for the number of components of $\Gamma[D_\Gamma]$ and enumerating them by $D_1, \dots, D_{m_\Gamma}$, the Gallai–Edmonds theorem [13, 17] states that

(i) every D_i is factor-critical;

(ii) $\Gamma[C_\Gamma]$ has a perfect matching;

(iii) every maximum matching of Γ consists of a nearly-perfect matching in each D_i, a perfect matching of $\Gamma[C_\Gamma]$, and $|A_\Gamma|$ edges that match the vertices of A_Γ to pairwise distinct components D_i.

In particular,

$$\mathrm{def}(\Gamma) = m_\Gamma - |A_\Gamma|.$$

Moreover, there is no edge of Γ between D_Γ and C_Γ. We refer to [24, Chapter 3] for the theorem and further background.

3 Proof outline

Deterministic setting: Theorem 1. If one base graph has a perfect matching, then automatically the product has a perfect matching. However, even if none of the base graphs has a perfect matching, the proportion of vertices covered by a maximal matching is preserved and we hope to increase it when taking products of higher and higher dimensions. We prove this by using that the edge-connectivity of a Cartesian product graph indeed increases with its dimension. With this at hand, let $U \subseteq V$ be an obstruction to a perfect matching in the sense of the Tutte-Berge formula, that is $G - U$ contains more than $|U|$ odd components. The edge-connectivity from Theorem 4.1 implies that the edge-boundary of U is of size at least $d - 1$, and a parity argument shows then shows that in fact this is of size least d. All these boundary edges must have one vertex in U and therefore, summing over the odd components in $G - U$, there can be at most $|U|$ of them since the graph under consideration is d-regular. By the Tutte-Berge formula, we conclude that such a product of sufficiently high dimension contains a perfect matching deterministically.

Random setting: Connectivity We prove a statement that holds simultaneously for every G_p with $p \in [p_-, p+]$. In order to treat the random graph G_p we use a classical sprinkling argument, that is, we expose the random edges in two independent rounds. The second, much sparser round, is referred to as the sprinkling step. After the first round, first-moment calculations yield the following properties of the resulting graph whp: its isolated vertices are at distance at least three in G and it has no component whose order lies in between 2 and d^{20}, see Theorems 5.1 and 5.2. It remains to merge the components of order at least d^{20}. For every partition of these components into two non-empty parts, the expansion properties of G yield many short edge-disjoint paths from one part to another. The probability that the second round contains none of these paths is very small, and we conclude using a union bound argument. This proves that whp in the range of $p \in [p_-, p+]$ there is a unique giant component and all vertices outside are isolated. Combining this with the fact that these vertices are at distance at least three in G, this implies the hitting time result for connectivity and minimum degree one. Indeed, each such isolated vertex joins this giant component once it receives an edge, and the isolated vertices disappear within that range of parameters whp. Thus, the graph becomes connected exactly at the same time the last isolated vertex receives an edge whp.

Random setting: perfect matching Again, we prove a statement that holds simultaneously for every G_p with $p \in [p_-, p+]$ under the additional assumption that $\delta(G_p) \geq 1$. By the first part, we my assume that such a random graph is connected whp. Our claim is that at this point it also contains a perfect matching. Suppose otherwise and consider the Edmond-Gallai decomposition (D, A, C) of G_p. If $D_1, ..., D_m$ are the components of $G_p[D]$ and writing $a := |A|$, then

$$m - a = def(G_p) \geq 2$$

since G has even order, so that every maximum matching leaves at least two vertices unmatched. The components D_i are odd, every component of $G_p[C]$ is even and there is no edge between D and C. Consequently,

$$odd(G_p - A) = m \geq a + 2$$

and A is an obstruction to a perfect matching in the sense of the Tutte-Berge formula. Our main task is thus to prove the following two statements:

1. Whp, there are no small sets U forming such an obstruction.
2. Whp, the set A from the Edmonds-Gallai decomposition of G_p is small.

For the first statement, we first prove any such minimal set U has neighbours in at least three odd components of $G_p - U$, see Theorem 6.3. Then we crucially use the expansion properties of the graph to show that a large number of edges must be absent in G_p to allow for such an obstruction U. This is combined with a union bound over the number of such obstructions depending on its size and expansion. Crucially, we prove an improved bound on the number of sets with small edge-boundary, see Theorem 6.4. This bound makes an explicit use of the Cartesian product structure: indeed, sets with bad expansion leave a 'fingerprint' on many coordinates, limiting their number.

It remains to study the Edmonds-Gallai decomposition of G_p to conclude that whp the part A is not too large. Again, note that every edge joining different components of $G_p[D]$ and every edge from D to

C must be absent in G_p. A degree count expresses the number of these absent edges in terms of the deficiency $def(G_p)$, the edges inside A, the boundary of C and the edge boundaries of components of $G_p[D]$, see (23). Using expansion properties of the graph, we lower-bound the number of edges leaving these components.

We then distinguish between the case when G is bipartite and non-bipartite. If G is bipartite, then every factor-critical bipartite graph consists of a single vertex, so all components of $G_p[D]$ are singletons. Using that the bipartition is balanced, we find a specific set whose internal edges must be absent and counting via choosing rooted spanning trees allows for a union bound.

If G is non-bipartite, some component of $G_p[D]$ could contain more than one vertex. Thus, we distinguish between the singleton and non-singleton components of D. After these components are fixed, it remains to choose the vertices in A and the set C. It remains crucial to balance a lower bound on the number of absent edges with an upper bound on the number of choices for such a structure. In particular, in Theorem 6.11, we bound the number of choices for a bipartition of a large vertex set with few internal edges in G_p.

Recalling the coupling of the random graph process with G_p, we note that all bounds require only that absent edges have labels larger than p_- and present edges have labels smaller than p^+. Therefore, they hold simultaneously throughout the interval $p \in [p_-, p^+]$, which contains the hitting time for minimum degree one (and connectivity) whp. The fact that at this point G_p also has a perfect matching shows the equality of the three hitting times.

4 Deterministic setting: Existence of a perfect matching

We will deduce the existence statement in Theorem 1 from edge-connectivity properties of the Cartesian product.

Lemma 4.1. *Let $G = \square_{i=1}^t H_i$, where every H_i is connected and d_i-regular. Let $M_i = |V(H_i)|$ and $d = \sum_{i=1}^t d_i$. Then*

$$\lambda(G) = \min\left\{ d,\ \min_{i\in[t]} \lambda(H_i) \prod_{j\neq i} M_j \right\}.$$

Proof. The displayed identity is the standard edge-connectivity formula for Cartesian products; see [18, Chapter 5]. □

We will use the following lemma, which follows almost immediately from the Tutte–Berge formula; we include its proof for the sake of completeness.

Lemma 4.2. *Let $d \geq 1$ be an integer and let Γ be a connected d-regular graph. Suppose that $\lambda(\Gamma) \geq d-1$. Then Γ has a perfect matching when $|V(\Gamma)|$ is even and a nearly-perfect matching when $|V(\Gamma)|$ is odd.*

Proof. For $U \subseteq V(\Gamma)$, let $Q_1, \ldots, Q_q$ be the odd components of $\Gamma - U$ such that $q = \mathrm{odd}(\Gamma - U)$. If $U = \varnothing$, then $q = 0$ when $|V(\Gamma)|$ is even and $q = 1$ when it is odd. Suppose now that $U \neq \varnothing$. Every Q_i is a proper non-empty set, so by the assumption that $\lambda(\Gamma) \geq d - 1$,

$$e_\Gamma(Q_i, Q_i^c) \geq d - 1.$$

Since $|Q_i|$ is odd and Γ is d-regular (implying that $d|Q_i| = 2e_\Gamma(Q_i) + e_\Gamma(Q_i, Q_i^c)$), we have

$$e_\Gamma(Q_i, Q_i^c) \equiv d|Q_i| \equiv d \pmod 2.$$

The cut has the same parity as d, whereas $d - 1$ has the opposite parity; hence

$$e_\Gamma(Q_i, U) = e_\Gamma(Q_i, Q_i^c) \geq d.$$

Summing over the odd components yields

$$qd \leq \sum_{i=1}^{q} e_\Gamma(Q_i, U) \leq d|U|,$$

and therefore $q \leq |U|$. The Tutte–Berge formula (4) implies that all vertices are covered by a maximal matching if $|V(\Gamma)|$ is even and all but at most one vertex are covered by a maximal matching if $|V(\Gamma)|$ is odd. □

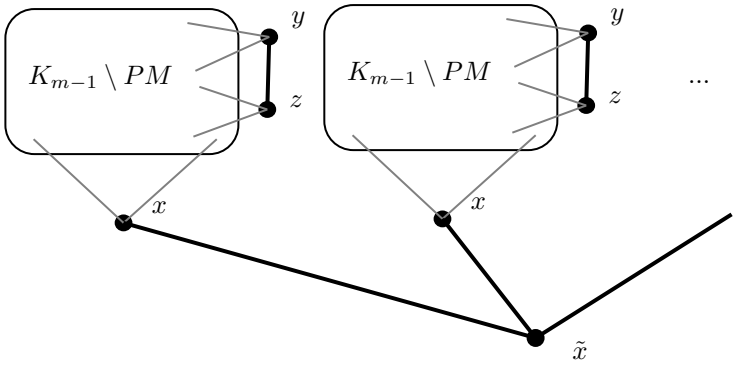


Figure 1: The construction of H.

Combining these two lemmas yields a perfect matching in product graphs of sufficiently high dimension.

Proof of Theorem 1. Let $G = \square_{i=1}^{t} H_i$. If a base graph H_i has a perfect matching M_i, then for each fixed choice of all coordinates other than the i-th, take the corresponding copy of M_i. The resulting matchings are vertex-disjoint and cover $V(G)$.

Otherwise, any base graph H_i has at least three vertices. Thus, by Theorem 4.1 we obtain that

$$\lambda(G) = \min\left\{d,\ \min_{i\in[t]} \lambda(H_i) \prod_{j\neq i} |V(H_j)|\right\} \geq \min\{d, 3^{t-1}\}. \tag{5}$$

Using that $d \leq t(M-1)$ and $t > 2\ln M$ we obtain that (5) is at least $d-1$. Applying Theorem 4.2, we obtain a (nearly-) perfect matching. This concludes the first part of the theorem.

For the second part, we provide an explicit construction where for $t < \ln M/3$ there is no perfect matching. Indeed, let M be sufficiently large, and let m be the largest odd integer with $(m+1)^2 \leq M$. Construct a graph H' as follows. Consider K_{m-1} and three distinct vertices x, y, and z. Remove one perfect matching from K_{m-1}, draw the edge yz, and draw all edges between x, y, z and K_{m-1}. Thus the vertex x has degree $m-1$, and every other vertex in H' has degree m. We then construct H by taking m copies of H' and a distinct vertex $\tilde{x}$, and drawing the edges between $\tilde{x}$ and the copy of x in each copy of H', see Figure 1. Thus H is an m-regular graph on $(m+1)^2$ vertices. Moreover, since m is odd, $H \setminus \{\tilde{x}\}$ has m odd components, and thus H has no perfect matching by Tutte's Theorem. Finally, construct G by taking the Cartesian product of H with $(t-1)$ triangles K_3. The base graphs all have size at most M, are connected and regular. Consider the set $U = \{\tilde{x}\} \times V(\square_{i=1}^{t-1} K_3)$. $G - U$ has exactly m odd components, so if $|U| = 3^{t-1} < m \leq \sqrt{M}$, G has no perfect matching. In particular, any $t < \ln M/3$ suffices. □

5 Connectivity in the random graph process

Throughout this section, let p satisfy

$$d^{-1/50} \leq n(1-p)^d \leq d^{1/50}. \tag{6}$$

We will expose the random edges in two independent rounds. In the first round every edge is present with probability p_1, and in the second round every edge is present with probability $p_2 = d^{-2}$, where $(1-p_1)(1-p_2) = 1-p$.

Note that (6) includes $p = p_-$ and $p = p_+$, as well as p_1 from the two-round exposure. In view of $d = \Theta_M(\ln n)$, (6) implies that both p and $1-p$ are bounded below by positive constants depending only on M. We begin by showing that isolated vertices are typically far apart in G.

Lemma 5.1. *Whp, every two isolated vertices of G_p are at distance at least three in G.*

Proof. There are at most nd ordered adjacent pairs and at most nd^2 ordered pairs at distance two. If distinct vertices u, v are not adjacent, the sets of edges incident with them are disjoint, and hence

$$\mathbb{P}(u, v \text{ are isolated in } G_p) = (1-p)^{2d}.$$

If u and v are adjacent, their incident edge sets overlap in one edge, so this probability is

$$(1-p)^{2d-1} \le \alpha_M (1-p)^{2d},$$

for some constant α_M. The expected number of ordered pairs of isolated vertices at distance at most two is therefore at most

$$\alpha_M n d^2 (1-p)^{2d} \le \frac{\alpha_M d^{2+1/25}}{n} = o(1),$$

where we used that p satisfies (6). We conclude with Markov's inequality. □

Let us further show the following gap statement, which follows from the expansion estimates in Theorem 2.1 and the rooted-tree bound in Theorem 2.2.

Lemma 5.2. *Whp, G_p has no component whose order lies in $[2, d^{20}]$.*

Proof. Fix $2 \le k \le d^{20}$. For every set $K \subseteq V$ of size k, the expansion from (3) gives

$$e_G(K, K^c) \ge k\big(d - (M-1)\log_M k\big) \ge 0.9kd.$$

If K is a component of G_p, it contains a spanning tree all of whose edges are present, while every edge of $E_G(K, K^c)$ is absent. Thus, using Theorem 2.2, we obtain

$$\mathbb{P}(G_p \text{ has a component of order } k) \le n(ed)^{k-1}(1-p)^{0.9kd} \le n^{1-0.8k},$$

Where the last inequality follows since $2 \le k \le d^{20}$, $n(1-p)^d \le d^{1/50}$ and $d = \Theta_M(\ln n)$. Summing over k shows that whp there is no component whose order lies in $[2, d^{20}]$. □

We are now ready to prove the key connectivity statement.

Proposition 5.3. *Whp, G_p has a unique component containing every non-isolated vertex. Its isolated vertices are pairwise at distance at least three in G.*

Proof. Let $p_2 = d^{-2}$, and choose p_1 so that

$$(1-p_1)(1-p_2) = 1-p.$$

The union of independent copies of G_{p_1} and G_{p_2} has the distribution of G_p, and

$$n(1-p_1)^d = \frac{n(1-p)^d}{(1-d^{-2})^d} = (1+o(1))n(1-p)^d.$$

The proofs of Theorems 5.1 and 5.2 hold in the extended regime given in (6) and thus they apply to G_{p_1}. Also, the expected number of isolated vertices in G_{p_1} is at most $2d^{1/50}$, and so Markov's inequality shows that fewer than $n/2$ vertices are isolated, and hence that there is a non-isolated component, with probability $1-o(1)$.

Using Theorems 5.1 and 5.2 we may condition on the event that every component of G_{p_1} is either an isolated vertex or has order at least d^{20}, and that the isolated vertices are pairwise at distance at least three in G. We will show that after sprinkling with p_2, typically all the large components, i.e., the components of order at least d^{20} merge. Let L be the union of the components of order at least d^{20}. Consider a non-trivial partition $A \sqcup B = L$ that respects these components and, by interchanging A and B, assume that $a := |A| \le |B|$. Put

$$A' = (A \cup N_G(A)) \setminus B, \qquad B' = V \setminus A'.$$

Both A' and B' have size at least a. Indeed, they contain A and B, respectively. Let D be the smaller of A' and B'. Then $a \le |D| \le n/2$, and the expansion from (2) gives

$$e_G(A', B') = e_G(D, D^c) \ge \frac{e}{M}|D| \ln \frac{n}{|D|} \ge \alpha_M a,$$

where α_M is a constant depending solely on M. For the last inequality, use the concavity of $x \mapsto x\ln(n/x)$ on $[a, n/2]$ and evaluate it at the endpoints.

For each edge of $E_G(A', B')$, choose one path of length at most three from A to B containing that edge. To see this, note that $V \setminus L$ consists of isolated vertices of G_{p_1}, no two of which are adjacent in

G. An isolated vertex in A' has a neighbour in A. An isolated vertex in B' has no neighbour in A; all its G-neighbours lie in L because distinct isolated vertices are non-adjacent, and therefore it has a neighbour in B. A fixed edge of G belongs to at most $O(d^2)$ of the chosen paths, because the boundary edge that determines such a path is at distance at most two from it in the line graph of G; thus each path has at most three edges. Greedily retaining such a path and deleting all other paths that share an edge with it therefore leaves at least

$$\frac{\alpha_M a}{d^2}$$

edge-disjoint paths from A to B for some constant $\alpha_M > 0$. The probability that none of them is present when exposing the edges of the second round is at most

$$(1-p_2^3)^{\alpha_M a/d^2} \le \exp\left\{-\frac{\alpha_M a}{d^8}\right\}.$$

It remains to take a union bound over the partitions of the large components in G_{p_1}. There are at most $N := n/d^{20}$ large components. For a partition whose smaller union has size a, that union contains at most a/d^{20} components. Consequently, the number of possible partitions is at most

$$\sum_{j \le a/d^{20}} \binom{N}{j} \le \exp\left\{O\left(\frac{a \ln n}{d^{20}}\right)\right\}.$$

This bound follows from the standard binomial estimate when $a/d^{20} \le N/2$, and from $2^N \le \exp(2a/d^{20})$ otherwise. Since $\ln n = O_M(d)$, summing over $d^{20} \le a \le n/2$ gives

$$\sum_{a=d^{20}}^{n/2} \exp\left\{O_M\left(\frac{a}{d^{19}}\right) - \frac{\alpha_M a}{d^8}\right\} = o(1).$$

Thus all non-isolated components of G_{p_1} merge after exposing the edges of the second round.

Finally, an isolated vertex of G_{p_1} that receives an edge in the second round cannot join another such vertex, because these vertices are not adjacent in G. It therefore joins the merged component. The vertices that receive no sprinkled edge remain isolated and by Theorem 5.1 they are still of distance at least three. □

Lemma 5.4. *Whp,*

$$1 \le \mathrm{iso}(G_{p_-}) \le d \qquad \textit{and} \qquad \mathrm{iso}(G_{p_+}) = 0.$$

Proof. Let $Y_p = \mathrm{iso}(G_p)$. Then $\mathbb{E}Y_p = n(1-p)^d$. For $p = p_+$ this expectation is $1/\omega$, and Markov's inequality gives $Y_{p_+} = 0$ whp. For $p = p_-$, we have $\mathbb{E}Y_{p_-} = \omega$, and

$$\mathbb{P}(Y_{p_-} > d) \le \frac{\omega}{d} = o(1).$$

For the lower bound, expand the second moment. Non-adjacent pairs of vertices are simultaneously isolated with probability $(\omega/n)^2$. An adjacent pair of vertices has probability at most $\alpha_M(\omega/n)^2$, where α_M is a suitable constant. Since G has $nd/2$ edges,

$$\mathbb{E}Y_{p_-}^2 \le \omega + \omega^2 + O\left(\frac{d\omega^2}{n}\right) = \omega^2 + O(\omega).$$

Hence $\mathrm{Var}(Y_{p_-}) = O(\omega) = o(\omega^2)$, and Chebyshev's inequality gives $Y_{p_-} > 0$ whp. □

Combining Theorem 5.3 and Theorem 5.4 proves part (a) and (b) of Theorem 3.

6 A perfect matching in the random graph process

In order to prove the last part of Theorem 3 we shall prove that, with high probability, every G_p with $p \in [p_-, p_+]$ and $\delta(G_p) \ge 1$ has a perfect matching. Given the results in Section 5, every such G_p is connected whp: the isolated vertices of G_{p_-} attach to the unique non-trivial component as soon as they receive an edge.

We make use of the Tutte-Berge formula. In particular, the formula (combined with the assumption that G has even order) implies that the existence of a set $U \subseteq V$ with

$$\operatorname{odd}(G_p - U) \geq |U| + 2 \tag{7}$$

is the only obstruction to a perfect matching in G_p. Define

$$u_0 \coloneqq \frac{n}{d^{M^5/p_-}}.$$

The following two lemmas comprise our two crucial ingredients.

Lemma 6.1. *Whp, simultaneously for every $p \in [p_-, p_+]$ with $\delta(G_p) \geq 1$, the graph G_p has no set U with $1 \leq |U| \leq u_0$ satisfying (7).*

Lemma 6.2. *Whp, simultaneously for every $p \in [p_-, p_+]$ with $\delta(G_p) \geq 1$, if $\operatorname{def}(G_p) \geq 2$, then the set A in the Gallai–Edmonds decomposition of G_p has size at most u_0.*

The proof of these two lemmas will be the task of the remaining subsections. First, we show that they indeed complete the proof of Theorem 3.

Proof. Let $p \in [p_-, p_+]$. Recall that whp if $\delta(G_p) \geq 1$, then G_p is connected by part (a) and (b) of Theorem 3 and we condition on this event. Suppose that G_p has no perfect matching and let (D, A, C) be its Gallai–Edmonds decomposition. Let $D_1, \dots, D_m$ be the components of $G_p[D]$. Put $a = |A|$ and $r = \operatorname{def}(G_p) \geq 2$ (since $|V(G)|$ is even). The set A is non-empty: if $A = \varnothing$, the distinct D_i are components of G_p, contrary to connectedness and $r \geq 2$. Each D_i has odd order, every component of $G_p[C]$ has even order, and there is no edge of G_p between D and C. Consequently,

$$\operatorname{odd}(G_p - A) = m = a + r \geq a + 2,$$

so A violates Tutte's condition, i.e, it satisfies (7). If $a \leq u_0$, this contradicts Theorem 6.1. If $a > u_0$, it contradicts Theorem 6.2. □

6.1 Tutte obstructions: Proof of Theorem 6.1

We first state and prove several lemmas that will be of use when proving Theorem 6.1.

We start with an observation concerning the structure of minimal Tutte obstructions.

Lemma 6.3. *Let U be a minimal set of vertices satisfying (7). Then every vertex $v \in U$ has neighbours (in G) in at least three components of $G_p - U$.*

Proof. Suppose towards contradiction that there is a minimal set U of size u satisfying (7), with $\ell > u$ components in $G_p - U$ and containing a vertex v whose neighbourhood in G intersects at most two components of $G_p - U$.

Consider $U' = U \setminus \{v\}$. Clearly, $|U'| = u - 1$. Observe that since v is connected to at most two components in $G_p - U$, there are at least $\ell - 1 > u - 1 = |U'|$ components in $G_p - U'$. But then U' is an obstruction, contradicting the minimality of U. □

Typically, subsets of vertices in G expand well. For sets with small edge boundary, we use a fingerprint argument to obtain an improved bound on the number of such sets. This makes use of the Cartesian product structure of G.

Lemma 6.4. *Let $\frac{n}{d^{\ln^2 d}} \leq a \leq \frac{n}{2}$. Then, the number of sets $A \subseteq V$ of size a with $e(A, A^c) < a \ln^2 d$ is at most $\exp\left\{\frac{2a}{\ln d}\right\}$.*

Proof. Let $\mathcal{F}$ be the family of $A \subseteq V$ satisfying the conditions of the lemma.

For $i \in [t]$ and any $A \subseteq V$, let $E_i(A, A^C) \subseteq E(A, A^C)$ be the set of edges in $E(A, A^C)$ corresponding to a change in the i-th coordinate, and let $e_{i,A} \coloneqq |E_i(A, A^C)|$. Moreover, given $I \subseteq [t]$, let $e_{I,A} = \sum_{i \in I} e_{i,A}$. We say that A is *bad* with respect to a set of coordinates I, if $e_{I,A} < a \ln^2 d \cdot \frac{|I|}{t}$. Let $\mathcal{A}_I$ be the family of sets A which are bad with respect to some $I \subseteq [t]$. Note that for every fixed $m \in \mathbb{N}$, if $A \in \mathcal{F}$, then there is some I with $|I| = m$ such that $e_{I,A} < a \ln^2 d \cdot \frac{|I|}{t}$. Thus,

$$|\mathcal{F}| \leq \sum_{\substack{I \subseteq [t] \\ |I| = m}} |\mathcal{A}_I| \leq \binom{t}{m} \max_{\substack{I \subseteq [t] \\ |I| = m}} |\mathcal{A}_I|.$$

We now set $m = \log_2\left(\ln^5 d\right)$, and turn to estimate $|\mathcal{A}_I|$ for any $I \subseteq [t]$ with $|I| = m$.

For such an I and $v \in V$, let $G(I,v) \coloneqq \square_{i\in[t]\setminus I}\{v_i\}\square_{i\in I}H_i \subseteq G$ be the product graph where we fix the coordinates outside of I to agree with v. That is, $G(I,v)$ is a product graph of dimension $|I| = m$. Observe that $2^{|I|} \le |V(G(I,v))| \le M^{|I|}$, and that for every $v \neq u \in V$, $V(G(I,v))$ and $V(G(I,u))$ are either disjoint or identical. Thus, fixing I with $|I| = m$, there are at most $\frac{n}{2^m}$ different subgraphs $G(I,v)$, and their union is V. We say that A *intersects non-trivially* with $G(I,v)$ if $V(G(I,v)) \cap A \neq \varnothing$ and $V(G(I,v)) \setminus A \neq \varnothing$. Applying Theorem 4.1 to $G(I,v)$, we have that if A intersects non-trivially with $G(I,v)$, then $G(I,v)$ spans at least $d(G(I,V)) \ge |I| = m$ edges of $E(A,A^C)$. Thus, if $A \in \mathcal{A}_I$, we have that A intersects non-trivially at most $\frac{a\ln^2 d}{t}$ such subgraphs. Indeed, otherwise, there would be no I with $|I| = m$ and $e_{I,A} < a\ln^2 d \cdot \frac{m}{t}$. Therefore, a set $A \in \mathcal{A}_I$ contains at least

$$\frac{a - M^m\frac{a\ln^2 d}{t}}{M^m} \le \frac{a}{M^m}$$

subgraphs of the form $G(I,v)$ for $v \in V$, and at most $M^m\frac{a\ln^2 d}{t}$ other vertices. We thus obtain that

$$|\mathcal{A}_I| \le \binom{\frac{n}{2^m}}{\frac{a}{M^m}}\binom{n}{M^m\frac{a\ln^2 d}{t}}, \tag{8}$$

where we first choose from the at most $n/2^m$ many the subgraphs $G(I,v)$ that are contained in A and then the other vertices. We may bound (8) from above using that $a \ge \frac{n}{d^{\ln^2 d}}$:

$$\left(\frac{enM^m}{a}\right)^{\frac{a}{M^m}}\left(\frac{ent}{M^m a\ln^2 d}\right)^{M^m\frac{a\ln^2 d}{t}} \le \left(ed^{\ln^2 d}M^m\right)^{\frac{a}{M^m}}\left(\frac{etd^{\ln^2 d}}{M^m\ln^2 d}\right)^{M^m\frac{a\ln^2 d}{t}}$$

$$\le \left(ed^{\ln^2 d}\ln^5 d\right)^{\frac{a}{\ln^5 d}}\left(\frac{ed^{\ln^2 d+1}}{\ln^7 d}\right)^{\frac{a\ln^7 d}{t}} \le \exp\left\{\frac{a}{\ln d}\right\}.$$

Altogether, by multiplying with the number of choices for I we obtain that

$$|\mathcal{F}| \le \binom{t}{\log_M(\ln^5 d)}\exp\left\{\frac{a}{\ln d}\right\} \le \exp\left\{\frac{2a}{\ln d}\right\},$$

as required. □

We set

$$u_1 \coloneqq \frac{n}{d^{M^3/p_-}}$$

and note that $u_1 \gg d\cdot u_0$.

Lemma 6.5. *Whp, there are no disjoint sets $U, B \subseteq V$ satisfying*

$$|U| \le u_0, \qquad u_1 \le |B| \le n/2,$$

for which every edge from B to $V\setminus(B\cup U)$ is not present in G_{p_-}.

Proof. Denote by $b \coloneqq |B|$. The number of possible U is at most

$$\sum_{u\le u_0}\binom{n}{u} \le \exp\{u_0\ln(en/u_0)\} = \exp\left\{u_0\cdot\frac{M^5}{p_-}\cdot\ln d\right\} \le \exp\left\{\frac{b}{\ln d}\right\},$$

where we used the definition of u_0 and u_1 and that $b \ge u_1$.

If $e_G(B,B^c) \ge b\ln^2 d$, then at least $b\ln^2 d - d|U| \ge b\ln^2 d/2$ edges from B avoid U. Thus, the probability that there are sets U, B with $b = |B|$ as in the claim is at most

$$\binom{n}{b}\exp\left\{\frac{b}{\ln d}\right\}(1-p_-)^{b\ln^2 d/2} = o(n^{-3}).$$

If $e_G(B,B^c) < b\ln^2 d$, Theorem 6.4 gives at most $\exp\{2b/\ln d\}$ choices for B. Moreover the expansion property in (2) gives $e_G(B,B^c) \ge \alpha_M b$ for some constant α_M while $du_0 = o(b)$, thus at least $\frac{\alpha_M}{2}b$ edges from B avoid U. Thus, the probability that there are sets U, B with $b = |B|$ as in the claim is at most

$$\exp\left\{\frac{3b}{\ln d}\right\}(1-p_-)^{\alpha_M b/2} \le \exp\{-\alpha'_M b\}.$$

A union bound over $b \ge u_1$ concludes the proof. □

Before turning to the key lemma of this subsection we give bounds on the expansion of the graph that will be useful throughout the calculations. Towards this, for $1 \le k \le \frac{n}{2}$ let

$$f^*(k) := \max\left\{k\big(d-(M-1)\log_M k\big), \frac{ek}{M}\ln\frac{n}{k}\right\}.$$

Extend it by $f^*(k) = f^*(n-k)$ for $n/2 < k < n$. By Theorem 2.1, every non-empty proper set $A \subseteq V$ satisfies

$$e_G(A, A^c) \ge f^*(|A|). \tag{9}$$

Lemma 6.6. *For all sufficiently large d and $3 \le k \le 2u_1$,*

$$f^*(k) \ge 2d + 15(M-1)k\ln d. \tag{10}$$

Proof. For $k = 3$, (9) together with the first term of (6.1) give

$$f^*(k) \ge 3d - 3(M-1)\log_M(3) \ge 2d + 45(M-1)\ln d,$$

as claimed. For $4 \le k \le d^3$ the same implies

$$f^*(k) \ge kd - 3k(M-1)\log_M(d) \ge 3d + 15k(M-1)\ln d.$$

If $d^3 < k \le 2u_1$, then (9) together with the second term of (6.1) give

$$f^*(k) \ge \frac{ek}{M}\left(\frac{M^3}{p_-}\ln d - \ln 2\right).$$

Using that $p_- \le 1 - 1/M$, we have $\frac{eM^2}{p_-} \ge 16(M-1)+1$. This yields

$$f^*(k) \ge 16(M-1)k\ln d + k\ln d - \frac{ek}{M}\ln 2 \ge 3d + 15(M-1)k\ln d,$$

where the last inequality uses that $k > d^3$. □

The following lemma is designed to show that whp there are no small sets U satisfying (7). Here, U plays the role of the obstruction and B contains components from $G_p - U$. Note that we treat components with an even number of vertices, except two, the same as components with an odd number of vertices, but this simplified point of view will be sufficient in this setting.

Lemma 6.7. *Whp, there are no $p \in [p_-, p_+]$ and disjoint $U, B \subseteq V$ with*

$$1 \le |U| \le u_0, \qquad |B| < u_1$$

such that

(i) no component of B in $G_p - U$ has size two;

(ii) the number of components of B in $G_p - U$ is at least $|U| + 1$;

(iii) every isolated vertex of B in $G_p - U$ has a G_p-neighbour in U;

(iv) every vertex in U has at least two neighbours in distinct components of B in $G_p - U$;

(v) no edge of G_p joins B to $V \setminus (U \cup B)$.

Proof. Denote by $u := |U|$ and by $b := |B|$. Let $Q_1, ..., Q_\ell$ be the components of B containing a single isolated vertex in $G_p - U$, and let $Q_{\ell+1}, ..., Q_{\ell+r}$ be the components of B size at least three in $G_p - U$ (see Figure 2). Then $\ell + r$ is the number of components of B in $G_p - U$. Further, we set $s = \sum_{i=\ell+1}^{\ell+r} |Q_i|$ to be the number of vertices contained in components of size at least three. On $U \cup B$, define a subgraph J of G by

$$E(J) = E_G(B) \cup E_G(U, B).$$

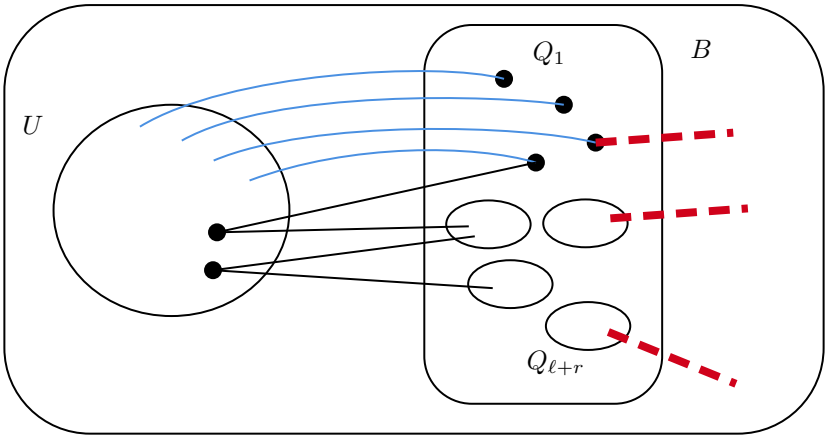


Figure 2: Sets U, B as in Theorem 6.7. All components of B in $G_p - U$ are either isolated vertices or of size at least three. Isolated vertices of B in $G_p - U$ have a G_p-neighbour in U (blue edges) and vertices in U have G_p-neighbours in at least two distinct components of B in $G_p - U$ (black edges). The dashed red edges are not present in G_p.

By (iii) and (iv) every component $K_1, \dots, K_a$ of J in G has order at least three, and hence $a \le \frac{b}{3}$. By assumption all edges between Q_i and Q_j for $i, j \le \ell + r$ with $i \ne j$ are not present in G_{p_-}. Furthermore, by (iv), any edge from B to $V \setminus (U \cup B)$ is not present in G_{p_-}. Denote by h the total number of these edges. Then,

$$2h \ge d\ell + \sum_{i=\ell+1}^{r} e_G(Q_i, Q_i^c) + \sum_{j=1}^{c} e_G(K_j, K_j^c) - du.$$

Indeed, every isolated vertex from a component Q_i for $i \le \ell$ has degree d in G and every other component Q_i for $\ell + 1 \le i \le \ell + r$ contributes $e_G(Q_i, Q_i^c)$ edges. With this, we have counted each edge between Q_i and Q_j for $i, j \le \ell + r$ with $i \ne j$ exactly twice, and the edges between B and $V \setminus (U \cup B)$ at most once. Adding the edges from $e_G(K_i, K_i^c)$ counts each edge at most twice. In the first two terms, we may have also counted edges between U and B (each such edge at most once), and in the second term we may have counted edges spanned by U (at most twice); all of these are accounted for when we subtract du.

Now, we use Theorem 6.6 (noting that all the components under consideration are of size at most $2u_1$) to give a lower bound on the number of absent edges. Using the expansion from (10), we obtain

$$2h \ge d(\ell - u) + 2dr + 15s(M-1)\ln d + 2da + 15(b+u)(M-1)\ln d.$$

Now $\ell + r \ge u + 1$, so $\ell - u + 2r \ge 1$ and consequently

$$h \ge da + \frac{d}{2} + \frac{15(M-1)}{2}(b + u + s)\ln d.$$

It remains to bound the number of choices for U, B satisfying the conditions. Fix the number a of components of J and $b + u$. First, we choose a root for each component and then a rooted spanning tree marking each of its vertices as belonging to U or B. These choices determine U, B and the component structure of B in $G_p - U$ and using Theorem 2.2 there are at most

$$n^a (ed)^{b+u} 2^{b+u} \le n^a d^{2(b+u)}$$

possibilities. Combining with a union bound over the number of choices for $b + u$ and a, we conclude that the probability to have sets B, U as in the claim is at most

$$\sum_{b+u=3}^{2u_1} \sum_{a \le (b+u)/3} n^a d^{2(b+u)} (1 - p_-)^{da + \frac{d}{2} + \frac{15(M-1)}{2}(b+u)\ln d}. \tag{11}$$

Using that $(1 - p_-)^d = \frac{\omega}{n}$ and $\ln n \ge \frac{\ln 2}{M-1} d$, the probability in (11) is bounded by

$$\left(\frac{\omega}{n}\right)^{1/2} \sum_{j=3}^{2u_1} j \left(d^2 \omega^{1/3} \left(\frac{\omega}{n}\right)^{\frac{15(M-1)}{2}\frac{\ln d}{d}} \right)^j = \left(\frac{\omega}{n}\right)^{1/2} \sum_{j \ge 3} j \exp\left(-\Omega(j \ln d)\right) = o(1),$$

since $\frac{15\ln 2}{2} \geq 5$.

□

We are now ready to derive Lemma 6.1.

Proof of Theorem 6.1. Recall that whp every G_p under consideration is connected and that G has even order. In particular, the empty set does not satisfy (7). Choose a set U satisfying (7) of minimum cardinality, put $u = |U|$, and let $K_1, \dots, K_m$ be the odd components of $G_p - U$. By Theorem 6.3 every $v \in U$ has neighbours in at least three of these components. Every K_i which is an isolated vertex in $G_p - U$ has a G_p-neighbour in U, because $\delta(G_p) \geq 1$.

Suppose that $u \leq u_0$. Whp, at most one K_i has order larger than u_1 by Theorem 6.5. Let B be the union of the remaining odd components. Note that there are at least $u+1$ of them and every $v \in U$ has neighbours in at least two components of B. If $|B| < u_1$, then U and B satisfy the assumptions of Theorem 6.7, a contradiction. If $|B| \geq u_1$, then a greedy union of its components has size between u_1 and $2u_1 < n/2$. In this case U together with this union of components satisfy the assumptions of Theorem 6.5, a contradiction. □

6.2 Edmonds-Gallai decomposition: Proof of Theorem 6.2

Fix $p \in [p_-, p_+]$ with $\delta(G_p) \geq 1$, and suppose that G_p has no perfect matching. Recall, that whp G_p is connected. We use the notation set out in Section 2 and abbreviate

$$D = D_{G_p}, \qquad A = A_{G_p}, \qquad C = C_{G_p}.$$

Let $D_1, \dots, D_m$ be the components of $G_p[D]$. Set

$$a = |A|, \qquad r = m - a = \operatorname{def}(G_p) \geq 2.$$

Here $r \geq 2$ because G_p has even order and no perfect matching. We start with some structural observations about the Edmonds-Gallai decomposition.

Let Q be the auxiliary bipartite graph with parts A and $\{D_1, \dots, D_m\}$, where $v \in A$ is adjacent to D_i whenever v has a G_p-neighbour in D_i.

Lemma 6.8. *For every non-empty $S \subseteq A$,*

$$|N_Q(S)| \geq |S| + 1.$$

Consequently every vertex of A has neighbours in at least two components of $G_p[D]$, and Q has at most r connected components.

Proof. For every i and every $v \in D_i$, the definition of D gives a maximum matching M_v leaving v unmatched; in particular, in M_v there is no edge from A to D_i. The edges in M_v match A to distinct members of $\{D_j : j \neq i\}$. Thus $Q - \{D_i\}$ has a matching saturating A. In particular, Q has a matching saturating A, so Hall's theorem gives $|N_Q(S)| \geq |S|$. If $|N_Q(S)| = |S|$, deleting any $D_i \in N_Q(S)$ would violate Hall's condition, a contradiction.

In each component of Q, the number of components of $G_p[D]$ is therefore at least one more than the number of vertices in A. Summed over all components of Q, this difference is $m - a = r$, so Q has at most r components. □

Let

$$h := \sum_{i<j} e_G(D_i, D_j) + e_G(D, C).$$

and note that all edges counted in h are not present in G_p (and each edge is counted once). We start by giving a lower bound on h.

Lemma 6.9.

$$2h = dr + 2e_G(A) + e_G(A \cup D, C) + \sum_{i:|D_i|\geq 3} \big(e_G(D_i, D_i^c) - d\big). \tag{12}$$

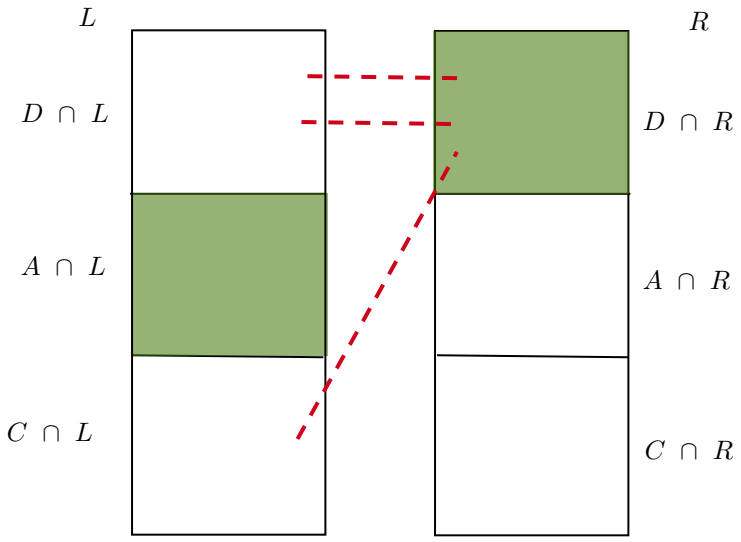


Figure 3: The setup for Theorem 6.10. The set S is depicted in green while the set D^* is depicted in stripes. The dashed red edges are not present in G_p.

Proof. Degree summation in D gives

$$d|D| = 2\sum_i e_G(D_i) + 2\sum_{i<j} e_G(D_i, D_j) + e_G(D, A) + e_G(D, C).$$

Rearranging for $h = \sum_{i<j} e_G(D_i, D_j) + e_G(D, C)$, we obtain

$$2h = d|D| - 2\sum_i e_G(D_i) - e_G(D, A) + e_G(D, C). \tag{13}$$

Similarly, degree summation in A gives

$$d|A| = 2e_G(A) + e_G(A, D) + e_G(A, C). \tag{14}$$

Subtracting the LHS and adding the RHS of (14) to (13), we obtain

$$2h = d(|D| - |A|) - 2\sum_i e_G(D_i) + 2e_G(A) + e_G(A, C) + e_G(D, C).$$

Now

$$|D| - |A| = r + \sum_{|D_i|\geq 3} (|D_i| - 1), \qquad 2e_G(D_i) = d|D_i| - e_G(D_i, D_i^c),$$

and since each $|D_i|$ is odd both sums over i may be combined, proving the claim. □

We first deal with the case where G is bipartite and $G_p[D]$ contains only isolated vertices.

Lemma 6.10. *Suppose that G is bipartite. Whp, for every $p \in [p_-, p_+]$ with $\delta(G_p) \geq 1$, there is no Gallai–Edmonds decomposition of G_p for which $\operatorname{def}(G_p) \geq 2$ and every component of $G_p[D]$ is an isolated vertex.*

Proof. Let $V = L \sqcup R$ be the bipartition of G and note that since G is regular, $|L| = |R|$. Since $G_p[C]$ has a perfect matching,

$$|C \cap L| = |C \cap R|.$$

Let

$$S = (A \cap L) \cup (D \cap R), \qquad D^* = D \cup (C \cap L),$$

and suppose w.l.o.g. that $|(A \cap L) \cup (D \cap R)| \leq |(A \cap R) \cup (D \cap L)|$ (otherwise set $S = (A \cap R) \cup (D \cap L)$ and $D^* = D \cup (C \cap R)$). Since L is an independent set in G, naturally so is $C \cap L$. Further, since there are no edges between distinct components of $G_p[D]$ and we assume all components of $G_p[D]$ are singletons,

there are no edges in $G_p[D]$. Finally, there are no edges in G_p between D and C by construction. Thus, every edge of $G[D^*]$ is not present in G_p. Moreover,

$$|D^*| - |V \setminus D^*| = r. \tag{15}$$

On the other hand,

$$e_G(D^*) + e_G(V \setminus D^*) = e_G(S, S^c),$$

see Figure 3. Regularity and (15) also give

$$2\big(e_G(D^*) - e_G(V \setminus D^*)\big) = dr.$$

Combining these two equations gives the identity

$$e_G(D^*) = \frac{dr}{4} + \frac{1}{2}e_G(S, S^c). \tag{16}$$

Consider the auxiliary graph Q and note that by Theorem 6.8 every vertex in $A \cap L$ has neighbours in at least two components of $D \cap R$. Furthermore, since G_p is whp connected every component of $D \cap R$ is whp connected to at least one vertex from $A \cap L$. We assume this holds deterministically, noting that this intersection changes the error probability only by a multiplicative factor $1 + o(1)$. Thus, every component of S in G_p has order at least three. Now, we turn to bounding the number of components ℓ of S in G_p. Note, that from the fact that every vertex in $A \cap L$ connects at least two components of $D \cap R$, we obtain

$$\ell \le |D \cap R| - |A \cap L|.$$

Now, since $r = |D \cap L| + |D \cap R| - |A \cap L| - |A \cap R|$ and our assumption that $|(A \cap L) \cup (D \cap R)| \le |(A \cap R) \cup (D \cap L)|$ we conclude that $\ell \le r/2$, i.e., that there are at most $r/2$ connected components in S. By (16), at least

$$\frac{dr}{4} + \frac{1}{2}e_G(S, S^c)$$

edges are not present in G_p. We distinguish a few cases according to S and in each case we bound the probability of the event in the claim by $(1-p_-)^{\frac{dr}{4}+\frac{1}{2}e_G(S,S^c)}$ and union bound over the number of choices for S.

If $|S| =: s \le 2u_1$, then the expansion from (10) and the fact that each component has order at least three give

$$e_G(S, S^c) \ge 2d\ell + 15(M-1)s\ln d.$$

By Theorem 2.2, there are most $n^\ell (ed)^s$ choices for S. Thus the probability to have such a decomposition is at most

$$\sum_{r\ge 2}\sum_{\ell\le r/2}\sum_{3\ell\le s\le 2u_1} n^\ell (ed)^s (1-p_-)^{\frac{dr}{4}+dc+7(M-1)s\ln d}. \tag{17}$$

Using that $(1-p_-)^d = \frac{\omega}{n}$ and $(1-p_-)^{7(M-1)s\ln d} \le d^{-2s}$ (since $\ln n \ge \frac{\ln 2}{M-1}d$ and $7\ln 2 \ge 2$), the probability in (17) is bounded by

$$\sum_{r\ge 2}\sum_{c\le r/2}\sum_{3c\le s\le 2u_1} \omega^c e^s d^{-s}\left(\frac{\omega}{n}\right)^{\frac{r}{4}} \le \sum_{r\ge 2} r\left(\frac{\omega^3}{n}\right)^{\frac{r}{4}} = o(1),$$

where we used that the sum over s is geometric and $c \le r/2$.

If $s > 2u_1$ and if $e_G(S, S^c) \ge s\ln^2 d$, then

$$\sum_s \binom{n}{s}(1-p_-)^{s\ln^2 d/2} = o(1),$$

since $s > 2u_1$ gives $\ln(en/s) = O_M(\ln d)$.

If $s > 2u_1$ and if $e_G(S, S^c) < s\ln^2 d$, then by Theorem 6.4 there are at most $\exp\{2s/\ln d\}$ choices for S. On the other hand, (2) gives

$$e_G(S, S^c) \ge \frac{e\ln 2}{M}s,$$

and hence

$$\sum_{s>2u_1} \exp\{2s/\ln d\}(1-p_-)^{e_G(S,S^c)/2} = o(1).$$

□

Next, we treat products with a non-bipartite base graph. We shall use the following counting lemma for bipartitions of vertex sets with few edges inside the bipartition classes. The two bipartition classes are labelled and either one may be empty.

Lemma 6.11. *Fix constants $K, K' > 0$. Let $R \subseteq V(G)$ have size $|R| \geq d$ and*

$$e_G(R, R^c) \leq K|R| \ln d.$$

The number of bipartitions $R = R_1 \sqcup R_2$ satisfying

$$e_G(R_1) + e_G(R_2) \leq K'|R|$$

is at most

$$\exp\left\{O\left(\frac{|R| \ln^2 d}{d}\right)\right\}.$$

Consequently, for every fixed $\theta > 0$,

$$\sum_{R=R_1 \sqcup R_2} (1-p_-)^{\theta(e_G(R_1)+e_G(R_2))} \leq \exp\{-o(|R|)\}. \tag{18}$$

Proof. Put $H = G[R]$ and $m = e_G(R_1) + e_G(R_2)$, and assume w.l.o.g. that $|R_2| \geq |R_1|$. Note that

$$e_G(R, R^c) = d|R| - \sum_{v \in R} d_H(v), \qquad 2m = \sum_{v \in R_1} d_{H[R_1]}(v) + \sum_{v \in R_2} d_{H[R_2]}(v). \tag{19}$$

Let X be the set of vertices $v \in R$ with either few neighbours in H or many neighbours in their own bipartition class, i.e.,

$$X \coloneqq \left\{v \in R : d_H(v) < \frac{3}{4}d \qquad \text{or} \qquad d_{H[R_i]}(v) > d/4 \quad (\text{ if } v \in R_i)\right\}.$$

By definition every $v \in R \setminus V$ has at least $\frac{d}{2}$ neighbours in the opposite bipartition class. From (19) we conclude that

$$|X| \leq \frac{4e_G(R, R^c)}{d} + \frac{8m}{d}. \tag{20}$$

Form a random set T by choosing each vertex of R_2 independently with probability $\rho = 12 \ln d/d$. The expected number of vertices of $R_1 \setminus X$ with no neighbour in T is at most

$$s(1-\rho)^{d/2} \leq \frac{|R|}{d^6}.$$

Moreover,

$$\mathbb{E}|T| = \rho|R_2| \leq \rho|R|, \qquad \mathbb{E}|R_2 \cap N_H(T)| \leq \rho 2e_G(R_2) \leq 2\rho m.$$

Applying Markov's inequality to these three quantities gives a set $T \subseteq R_2$ with

$$|T| \leq 3\rho|R|, \qquad |R_1 \setminus N_H(T)| \leq |X| + \frac{|R|}{d^2}, \qquad |R_2 \cap N_H(T)| \leq 6\rho m.$$

Combining (20) with our assumptions on $e_G(R, R^c)$ and m, we obtain that $R_1 \triangle N_H(T)$ has size $O(|R| \ln d/d)$. The pair $(T, R_1 \triangle N_H(T))$ determines R_1, and the number of possible pairs is at most

$$\left(\sum_{j \leq O(|R| \ln d/d)} \binom{|R|}{j}\right)^2 = \exp\left\{O\left(\frac{|R| \ln^2 d}{d}\right)\right\},$$

which proves the first assertion.

For (18) apply the first assertion for the bipartitions with $m \leq \frac{2(M-1)}{\Theta}|R|$ to obtain that these contribute at most $\exp\{o(|R|)\}$. For the bipartitions with $m > \frac{2(M-1)}{\Theta}|R|$ contribute at most

$$2^{|R|}(1-p_-)^{2(M-1)|R|} = \exp\left(-\Omega(|R|)\right),$$

using again that $\ln n \geq \frac{\ln 2}{M-1}d$. □

We can now show that if G is non-bipartite, then the set A in the Gallai-Edmonds decomposition of G_p cannot be too big.

Lemma 6.12. *Suppose that at least one base graph of G is non-bipartite. Whp for every $p \in [p_-, p_+]$, if $\operatorname{def}(G_p) \geq 2$, then the set A in the Gallai–Edmonds decomposition of G_p has size at most u_0.*

Proof. Fix $p \in [p_-, p_+]$, let (D, A, C) be the Gallai–Edmonds decomposition of G_p, and suppose that $\operatorname{def}(G_p) \geq 2$ and $|A| > u_0$. Let D_0 be the union of the singleton components of $G_p[D]$, let D^+ be the family of non-singleton components of $G_p[D]$, and put

$$D_1 = \bigcup_{D_i \in D^+} D_i.$$

Write $a = |A|$, $r = |D^+| + |D_0| - a \geq 2$, and recall

$$h = \sum_{i<j} e_G(D_i, D_j) + e_G(D, C).$$

All the edges counted in h are not present in G_{p_-}. Hence, we shall sum $(1-p_-)^h$ over the choices of A, C, the singleton components of D, and the non-singleton components of D.

Every $D_i \in D^+$ is a non-singleton factor-critical component, so $|D_i| \geq 3$. By (12),

$$h = \frac{dr}{2} + e_G(A) + \frac{e_G(C, C^c)}{2} + \frac{1}{2} \sum_{D_i \in D^+} \big(e_G(D_i, D_i^c) - d\big). \tag{21}$$

Also

$$h \geq e_G(D_0), \tag{22}$$

since every edge of $G[D_0]$ joins two distinct components of $G_p[D]$. Taking three quarters of (21) and one quarter of (22) gives

$$h \geq \frac{3dr}{8} + \frac{3}{4} e_G(A) + \frac{1}{4} e_G(D_0) + \frac{3}{8} e_G(C, C^c) + \frac{3}{8} \sum_{D_i \in D^+} \big(e_G(D_i, D_i^c) - d\big). \tag{23}$$

Put $k_i = \min\{|D_i|, |D_i^c|\}$ so that $3 \leq k_i \leq n/2$. If $k_i \leq 2u_1$, then (10) gives $e_G(D_i, D_i^c) \geq 2d$. If $k_i > 2u_1$, then $\ln(n/k_i) \geq \ln 2$, and (2) gives

$$e_G(D_i, D_i^c) \geq \frac{e \ln 2}{M} k_i > 2d$$

since $u_1 \gg d$. Thus,

$$\frac{1}{8}\big(e_G(D_i, D_i^c) - d\big) \geq \frac{1}{16} e_G(D_i, D_i^c). \tag{24}$$

Furthermore,

$$e_G(C, C^c) + \sum_{D_i \in D^+} e_G(D_i, D_i^c) \geq e_G((C \cup D_1), (C \cup D_1)^c). \tag{25}$$

Plugging (24) and (25) into (23) we obtain the following lower bound for h:

$$h \geq \frac{3dr}{8} + \frac{e_G(A) + e_G(D_0)}{4} + \frac{e_G((C \cup D_1), (C, \cup D_1)^c)}{16} + \frac{1}{4} \sum_{D_i \in D^+} \big(e_G(D_i, D_i^c) - d\big). \tag{26}$$

We first sum over the possible families D^+ for a fixed set $(C \cup D_1)$. By Theorem 2.2 here are at most $|C \cup D_1|(ed)^{k-1}$ connected sets of cardinality k with a root in $C \cup D_1$.

For $3 \leq |D_i| \leq 2u_1$, Theorem 6.6 gives

$$e_G(D_i, D_i^c) - d \geq d + 15(M-1)|D_i| \ln d.$$

Plugging this into the last term of (26) the total contribution from these sets is at most

$$\begin{aligned}
&|C \cup D_1| \sum_{k=3}^{2u_1} (ed)^{k-1} (1-p_-)^{\frac{d+15(M-1)k\ln d}{4}} \\
\leq & n^{3/4} \omega^{1/4} \sum_{k\geq 3} (ed)^{k-1} d^{-\frac{15\ln 2}{4}k} \\
\leq & n^{3/4+o(1)},
\end{aligned}$$

where we used that $\ln n \geq \frac{\ln 2}{M-1} d$ and that $\frac{15\ln 2}{4} > 1$.

For $2u_1 < |D_i| \leq n/2$, first suppose that

$$e_G(D_i, D_i^c) \geq \frac{5(M-1)}{\ln 2} |D_i| \ln d.$$

Using again $\ln n \geq \frac{\ln 2}{M-1} d$, the contribution from this range is at most

$$\begin{aligned}
& n(1-p_-)^{-d/4} \sum_{k>2u_1} (ed)^k (1-p_-)^{\frac{5(M-1)}{4\ln 2} k \ln d} \\
\leq & \left(\frac{\omega}{n}\right)^{-1/4} \sum_{k>2u_1} (ed)^k d^{-\frac{5}{4}k} \leq \exp\{-\Omega(u_1 \ln d)\}.
\end{aligned}$$

Indeed, $\frac{5}{4} > 1$ makes the series geometric with exponent $-\Omega(k \ln d)$, while the prefactor has logarithm $O(d)$ and $u_1 \gg d$.

Otherwise if $e_G(D_i, D_i^c) < |D_i| \ln^2 d$, then Theorem 6.4 (which is applicable because $2u_1 \geq \frac{n}{d^{\ln^2 d}}$) gives at most $\exp\{2|D_i|/\ln d\}$ choices, whereas

$$e_G(D_i, D_i^c) - d = \Omega(|D_i|)$$

since $|D_i| > 2u_1 \gg d$. Thus,

$$\begin{aligned}
& \sum_{k>2u_1} \exp\left\{\frac{2k}{\ln d}\right\} (1-p_-)^{\Omega(k)} \\
\leq & \sum_{k>2u_1} \exp\{-\Omega(k)\} = \exp\{-\Omega(u_1)\}.
\end{aligned}$$

There is at most one D_i with $|D_i| > n/2$. We now do a similar argument focusing on its complement D_i^c. Since $A \subseteq K^c$, we have $u_0 < |D_i^c| < n/2$. If

$$e_G(D_i, D_i^c) \geq \frac{8(M-1)}{\ln(2e)} |D_i^c| \ln(en/|D_i^c|),$$

then,

$$\sum_{u_0<k<n/2} \binom{n}{k} (1-p_-)^{\left\{\frac{8(M-1)}{\ln(2e)} k \ln(en/k) - d\right\}/4} = \exp(-\Omega(u_0)),$$

using that $\log n \geq \frac{\log 2}{M-1} d$ and $2\log 2 > 1$. Otherwise if $e_G(D_i, D_i^c) \leq |D_i^c| \ln^2 d$, then by Theorem 6.4 (which is applicable because $u_0 \geq \frac{n}{d^{\ln^2 d}}$) and $e_G(D_i, D_i^c) - d \geq \Omega(|D_i^c|)$. Here again $|D_i^c| > u_0 \gg d$ permits the subtraction of d from the linear expansion bound. Then,

$$\sum_{u_0<k<n/2} \exp\{2k/\ln d\} (1-p_-)^{\Omega(k)} = \exp(-\Omega(u_0)).$$

Let $\mathcal{S}_{\leq n/2}$ be the sum of the terms corresponding to connected sets of size at most $n/2$, and let $\mathcal{S}_{>n/2}$ be the corresponding sum for a possible connected set larger than $n/2$. The preceding estimates give

$$\mathcal{S}_{\leq n/2} \leq n^{3/4+o(1)}, \qquad \mathcal{S}_{>n/2} \leq e^{-\Omega_M(u_0)}.$$

A pairwise disjoint family contains at most one member larger than $n/2$. In particular, uniformly for sets Y with $|V \setminus Y| > u_0$, the sum over unordered families of pairwise disjoint connected subsets of $G[Y]$ satisfies

$$\sum_{\substack{D^+ \\ |V\setminus(C\cup D_1)|>u_0}} (1-p_-)^{\frac{1}{4}\sum_{D_i\in D^+}\{e_G(D_i,D_i^c)-d\}} \le \exp\{n^{3/4+o(1)}\} = \exp\{o(u_0)\}. \tag{27}$$

This sum counts only the non-singleton components of $G_p[D]$; it does not use the vertices or internal edges of C. Given $C\cup D_1$, an ordered partition $V\setminus(C\cup D_1) = A\sqcup D_0$, and the family D^+, the remaining set $C = (C\cup D_1)\setminus\bigcup_{D_i\in D^+} D_i$ is determined.

Recalling (26), after (27) and after discarding $(1-p_-)^{3dr/8} \le 1$, it remains to bound the following sum over ordered partitions $V = (C\cup D_1)\sqcup A\sqcup D_0$:

$$\sum_{\substack{V=(C\cup D_1)\sqcup A\sqcup D_0 \\ |A|>u_0}} (1-p_-)^{e_G((C\cup D_1),(C\cup D_1)^c)/16+\{e_G(A)+e_G(D_0)\}/4}. \tag{28}$$

We split according to whether $s := |(C\cup D_1)^c| \le n/2$. Suppose first that this inequality holds. Then $s \ge |A| > u_0$ and $\ln\left(\frac{en}{s}\right) = O(\ln d)$. If

$$e_G((C\cup D_1),(C\cup D_1)^c) \ge \frac{20(M-1)}{\ln 2} s\ln\left(\frac{en}{s}\right), \tag{29}$$

then

$$\binom{n}{s} 2^s (1-p_-)^{e_G((C\cup D_1),(C\cup D_1)^c)/16} \le \exp\{-\Omega(s\ln(en/s))\}.$$

If (29) fails, then

$$e_G((C\cup D_1),(C\cup D_1)^c) = O(s\ln d) < s\ln^2 d.$$

By Theorem 6.4, there are $e^{o(s)}$ choices for $(C\cup D_1)^c$, while expansion (Theorem 2.1) gives $e_G((C\cup D_1),(C\cup D_1)^c) = \Omega(s)$. For each fixed $(C\cup D_1)^c$, Theorem 6.11 gives

$$\sum_{A\sqcup D_0} (1-p_-)^{\{e_G(A)+e_G(D_0)\}/4} \le e^{o(s)},$$

obtained from (18) with $\theta = 1/4$.

Thus, the range $s \le n/2$ contributes at most

$$\sum_{s>u_0} e^{-\Omega(s)} = e^{-\Omega(u_0)}. \tag{30}$$

In particular, when the Gallai–Edmonds set $C\cup D_1$ contains more than half of vertices, we count the smaller set $A\cup D_0$ rather than $C\cup D_1$.

We may now assume $s = |(C\cup D_1)^c| > n/2$, so $|C\cup D_1| < n/2$. Fix $\varepsilon > 0$ to be chosen sufficiently small. Suppose first that $|C\cup D_1| \ge \varepsilon n$. If $e_G((C\cup D_1),(C\cup D_1)^c) \ge \frac{20(M-1)\ln 3}{\ln 2} n$, then

$$3^n (1-p_-)^{e_G((C\cup D_1),(C\cup D_1)^c)/16} \le e^{-\Omega(n)}.$$

Otherwise,

$$e_G((C\cup D_1),(C\cup D_1)^c) = O(n) = O(|C\cup D_1|) < |C\cup D_1|\ln^2 d.$$

By Theorem 6.4, there are $e^{o(n)}$ choices for $(C\cup D_1)^c$, and, for each fixed $(C\cup D_1)^c$, (18) with $\theta = 1/4$ (which is applicable since $e_G((C\cup D_1),(C\cup D_1)^c) < \frac{20(M-1)\ln 3}{\ln 2} n \le \frac{40(M-1)\ln 3}{\ln 2} |(C\cup D_1)^c|$) gives

$$\sum_{R=A\sqcup D_0} (1-p_-)^{\{e_G(A)+e_G(D_0)\}/4} \le e^{o(n)}.$$

Moreover,

$$e_G((C\cup D_1),(C\cup D_1)^c) = \Omega(|C\cup D_1|) \ge \alpha_M \varepsilon n,$$

for some suitable constant α_M. So, in total this contributes $e^{-\Omega(n)}$.

Finally suppose that $|C\cup D_1| < \varepsilon n$. Fix i such that H_i is non-bipartite. Fixing all coordinates other than the i-th partitions V into $n/|V(H_i)|$ sets, each of which induces a copy of H_i. At most $|C\cup D_1|$ of

these copies meet $C \cup D_1$. In every remaining copy, the vertices lie in $A \cup D_0$; since H_i is non-bipartite, at least one edge has both endpoints in A or both endpoints in D_0. Thus

$$e_G(A) + e_G(D_0) \geq n/M - |C \cup D_1| \geq n/(2M). \tag{31}$$

The ordered partitions for which $e_G(A) + e_G(D_0) \geq \frac{5(M-1)\log 3}{\log 2} n$ contribute at most

$$3^n (1-p_-)^{\frac{5(M-1)\log 3}{4\log 2} n} = e^{-\Omega(n)}.$$

For the remaining ordered partitions with $e_G(A)+e_G(D_0) < \frac{5(M-1)\log 3}{\log 2} n$, repeat the counting argument from Theorem 6.11 for the ordered partition $R = A \sqcup D_0$. The trivial bound gives $e_G((C \cup D_1), (C \cup D_1)^c) \leq d|C \cup D_1|$, so (20) gives only $\varepsilon n + O(n/d)$ exceptional vertices in X. As in the proof of Theorem 6.11, the random sample T, together with $A \triangle N_{G[R]}(T)$ after possibly interchanging A and D_0, determines the ordered partition of R. We get

$$|A \triangle N_{G[R]}(T)| \leq \epsilon n + o(n), \qquad |T| = o(n).$$

The number of choices for $(T, A \triangle N_{G[R]}(T))$ is at most

$$\exp\{\varepsilon \ln\left(\frac{e}{\varepsilon}\right) n + o(n)\}$$

and the choices of $(C \cup D_1)$ contribute at most $\exp\{\varepsilon \ln(e/\varepsilon) n\}$. Recalling that by (31) there are at least $(e_G(A) + e_G(D_0))/4 \geq n/(8M)$ edges that are not present in G_{p_-}. Choosing $\varepsilon = \varepsilon(M) > 0$ sufficiently small ensures $2\varepsilon \log\left(\frac{e}{\varepsilon}\right) < \frac{\log 2}{8M(M-1)}$. Then, the contribution is at most

$$\begin{aligned}
&\exp\{\varepsilon \ln\left(\frac{e}{\varepsilon}\right) n + o(n)\}(1-p_-)^{n/(8M)} \\
\leq &\exp\{2\varepsilon \ln\left(\frac{e}{\varepsilon}\right) n - \frac{\log 2}{8M(M-1)} + o(n)\} \\
= &\exp(-\Omega(n)).
\end{aligned}$$

Combining (30) with this contribution from $s > \frac{n}{2}$ proves that (28) is at most $\exp(-\Omega(u_0)) + \exp(-\Omega(n))$.

Multiplying (27) with this bound on (28) gives $o(1)$, showing that whp there is no Gallai–Edmonds decomposition of G_p with $\mathrm{def}(G_p) \geq 2$ and $|A| > u_0$. □

Lemma 6.2 will follow directly from Lemmas 6.10 and 6.12.

Proof of Theorem 6.2. If G is bipartite, every factor-critical bipartite graph is a singleton, so the claim follows from Theorem 6.10. If G has a non-bipartite base graph, it follows from Theorem 6.12. □

7 Discussion

The deterministic part identifies a natural sufficient condition in terms of the dimension of a product graph to contain a perfect matching that is asymptotically correct. We wonder whether other natural properties or spanning structures are guaranteed by a high dimension of Cartesian product graphs. In particular, we ask the following question for Hamilton cycles.

Question 7.1. *Let H_i for $i \in [t]$ be a sequence of regular graphs of bounded size. Is there a dimension t^* such that for all $t \geq t^*$ the Cartesian product graph $G = \square_{i=1}^t H_i$ has a Hamilton cycle?*

In the probabilistic setting we have extended the classical result concerning hitting times of minimum degree one, connectivity, and the existence of a perfect matching to random subgraphs of regular Cartesian product graphs. In particular, this includes a simplified self-contained version of the connectivity result for bond percolation on the hypercube. Let us mention that, independently, Collares, Doolittle, and Erde use a similar approach – that is, sprinkling with probabilities p_1 and p_2 – to show a connectivity result for bond percolation on the permutahedron [8]. There, however, similarly to the approach of [1, 5, 22], one utilises that in G_{p_1}, large components are relatively well-spread, that is, typically every vertex in G is quite close (in G) to a large component of G_{p_1}. In this paper, we neither require nor utilise such a 'density' statement, and instead use the fact that the isolated vertices are 'sparsely spread'.

Two features of the proof appear useful beyond this setting. First, degree summation over the Gallai–Edmonds decomposition turns the matching deficiency into a lower bound on the number of edges that must be absent, expressed through the boundaries of the components of $G_p[D]$. Second, the product structure of the graph allows to bound the number of large sets with a small edge-boundary much more efficiently, enabling the use of the union bound, see Theorem 6.4.

Many other random graph models are known to have typically the same hitting times for minimum degree one, connectivity, and the existence of a perfect matching (see, for example, [16] and the references therein). It is thus natural to ask what are the minimal requirements on G for this phenomenon to hold. As a step towards this, we propose the following question, considering regular graphs with high-degree.

Question 7.2. *Let G be a d-regular graph on n vertices, with $d = \omega(1)$ and n divisible by two. What minimal requirements are needed on G, such that in the random graph process on G, the hitting times for minimum degree one, connectivity, and the existence of a perfect matching are the same?*

Acknowledgements The authors would like to thank Joshua Erde, Mihyun Kang, and Michael Krivelevich for their guidance, advice, and fruitful discussions. Special thanks to an anonymous referee for spotting an error in a previous version. This research was funded in part by the Austrian Science Fund (FWF) [10.55776/W1230, 10.55776/F1002].

Statement about the use of AI The authors acknowledge the use of AI tools for polishing this paper, and note that all of the proof ideas were entirely human-generated.

References

[1] M. Ajtai, J. Komlós, and E. Szemerédi. Largest random component of a k-cube. *Combinatorica*, 2(1):1–7, 1982.

[2] Claude Berge. *The theory of graphs and its applications*. London: Methuen & Co. Ldt.; New York: John Wiley & Sons. Inc. X, 247 p. (1962)., 1962.

[3] A. Beveridge, A. Frieze, and C. McDiarmid. Random minimum length spanning trees in regular graphs. *Combinatorica*, 18(3):311–333, 1998.

[4] B. Bollobás. Complete matchings in random subgraphs of the cube. *Random Structures Algorithms*, 1(1):95–104, 1990.

[5] B. Bollobás, Y. Kohayakawa, and T. Łuczak. The evolution of random subgraphs of the cube. *Random Structures Algorithms*, 3(1):55–90, 1992.

[6] Béla Bollobás and Andrew Thomason. Random graphs of small order. In *Random graphs '83 (Poznań, 1983)*, volume 118 of *North-Holland Math. Stud.*, pages 47–97. North-Holland, Amsterdam, 1985.

[7] Ju. D. Burtin. The probability of connectedness of a random subgraph of an n-dimensional cube. *Problemy Peredači Informacii*, 13(2):90–95, 1977.

[8] M. Collares, J. Doolittle, and J. Erde. The evolution of the permutahedron. *arXiv:2404.17260*, 2024.

[9] S. Diskin, J. Erde, M. Kang, and M. Krivelevich. Isoperimetric inequalities and supercritical percolation on high-dimensional graphs. *Combinatorica*, 44:741–784, 2024.

[10] Sahar Diskin, Joshua Erde, Mihyun Kang, and Michael Krivelevich. Percolation on irregular high-dimensional product graphs. *Combin. Probab. Comput.*, 33(3):377–403, 2024.

[11] Sahar Diskin, Joshua Erde, Mihyun Kang, and Michael Krivelevich. Percolation on high-dimensional product graphs. *Random Structures & Algorithms*, 66(1):e21268, 2025.

[12] Sahar Diskin and Wojciech Samotij. Isoperimetry in product graphs. *Electron. J. Combin.*, 32(3):Paper 3.12, 2025.

[13] Jack Edmonds. Paths, trees, and flowers. *Canadian Journal of mathematics*, 17:449–467, 1965.

[14] P. Erdős and A. Rényi. On the existence of a factor of degree one of a connected random graph. *Acta Math. Acad. Sci. Hungar.*, 17:359–368, 1966.

[15] P. Erdős and J. Spencer. Evolution of the n-cube. *Comput. Math. Appl.*, 5(1):33–39, 1979.

[16] Alan Frieze and Boris Pittel. Perfect matchings in random graphs with prescribed minimal degree. In *Mathematics and computer science. III*, Trends Math., pages 95–132. Birkhäuser, Basel, 2004.

[17] Tibor Gallai. Maximale systeme unabhängiger kanten. *A Magyar Tudományos Akadémia Matematikai Kutató Intézetének Közleményei*, 9(3):401–413, 1964.

[18] Richard Hammack, Wilfried Imrich, and Sandi Klavžar. *Handbook of product graphs*. Discrete Mathematics and its Applications (Boca Raton). CRC Press, Boca Raton, FL, second edition, 2011. With a foreword by Peter Winkler.

[19] Felix Joos. Random subgraphs in Cartesian powers of regular graphs. *Electron. J. Combin.*, 19(1):Paper 47, 12, 2012.

[20] Felix Joos. Random subgraphs in sparse graphs. *SIAM J. Discrete Math.*, 29(4):2350–2360, 2015.

[21] Anton Kotzig. 1-factorizations of Cartesian products of regular graphs. *J. Graph Theory*, 3(1):23–34, 1979.

[22] Michael Krivelevich. Component sizes in the supercritical percolation on the binary cube. *arXiv preprint arXiv:2311.07210*, 2023.

[23] L. Lichev. The giant component after percolation of product graphs. *J. Graph Theory*, 99(4):651–670, 2022.

[24] László Lovász and Michael D. Plummer. *Matching Theory*, volume 121 of *North-Holland Mathematics Studies*. North-Holland, Amsterdam, 1986.

[25] A. A. Sapoženko. Metric properties of almost all functions of the algebra of logic. *Diskret. Analiz*, 10:91–119, 1967.